PAPER GRANNY
arXiv Paper Interpretation Agent

Gated Attention for Large Language Models

A Gentle but Deep Reading Report

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Paper Information

Original paper: arXiv:2505.06708

Title: Gated Attention for Large Language Models: Non-linearity, Sparsity, and Attention-Sink-Free

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1 Background Knowledge

What problem area is this paper in?

This paper studies the **attention layer** inside large language models (LLMs). Attention is the part that lets each token look at earlier tokens and decide which past information matters. If you ask a model to continue a sentence, attention is one of the main tools it uses to remember what came before.

A standard Transformer attention block does four things:

- it turns the hidden states into **queries**, **keys**, and **values**,
- it compares queries with keys to get attention weights,
- it uses those weights to mix the values,
- it projects the mixed result back into the model dimension.

The paper asks a surprisingly simple question:

If we insert a small gate into attention, where should we put it, and why does it help?

Plain-language meaning of "gate"

A gate is like a smart faucet or valve. It does not create information from scratch. Instead, it decides *how much of an existing signal is allowed to pass through*. If the gate value is close to 1, most information passes. If it is close to 0, that part is mostly suppressed.

Gates are common in older neural networks such as LSTMs and GRUs, and also in modern architectures such as SwiGLU and state-space models. But in standard softmax attention, people often use the mechanism without carefully separating *what exactly the gate itself contributes*. This paper tries to isolate that contribution.

What is an attention sink?

An **attention sink** means some token, often the very first token, receives too much attention again and again, even when it is not truly informative. A simple analogy: imagine a meeting where everyone keeps looking back at the chairperson for no real reason. The chairperson becomes an "attention sink" that absorbs too much focus.

The authors claim that a simple gate placed **after SDPA** (Scaled Dot-Product Attention) not only improves performance, but also reduces attention sinks, improves training stability, and helps long-context extrapolation.

2 Core Question and Main Innovation

One-sentence summary

The paper shows that **adding a head-specific sigmoid gate after the attention output of each head** is a very effective modification for LLMs, because it adds **non-linearity** and **input-dependent sparsity** to the attention pathway.

The authors test many variants, not just one. They vary:

- **position**: after query, key, value, SDPA output, or final output,
- **granularity**: headwise or elementwise,
- **sharing**: head-specific or head-shared,
- **operation**: multiplicative or additive,
- **activation**: sigmoid or SiLU.

This is important because it turns the paper from "we tried one trick and it worked" into "we compared many nearby possibilities and found the important ingredients." That makes the conclusions much more convincing.

The paper's main empirical conclusion

Among many tested variants, the best and most consistent one is:

apply a head-specific multiplicative sigmoid gate after SDPA output (G_1)

This improves perplexity, benchmark scores, training stability, and long-context behavior.

3 Method Details

3.1 Standard attention before gating

Let the input hidden states be $X \in \mathbb{R}^{n \times d_{\text{model}}}$. The standard attention first computes

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

Here:

- n is the sequence length,
- d_{model} is the hidden dimension,
- W_Q, W_K, W_V are learned matrices.

Then each query compares itself to keys:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V.$$

In plain English:

- QK^T says how relevant previous tokens are,
- softmax converts these numbers into nonnegative weights that sum to 1,
- those weights are used to average the values V.

Finally, different heads are concatenated and sent through an output projection:

$$O = \text{MultiHead}(Q, K, V)W_O.$$

3.2 How the gate is defined

The paper uses a general gate of the form

$$Y' = g(Y, X, W_\theta, \sigma) = Y \odot \sigma(XW_\theta).$$

This formula means:

- Y is the signal we want to modulate,
- XW_θ computes gate scores from the current hidden state,
- σ is an activation function such as sigmoid,
- $\odot$ means elementwise multiplication,
- Y' is the filtered output.

Intuition for the formula

Suppose Y is the information that attention wants to send forward. The gate computes a mask from the current token's hidden state. That mask says: "keep this part, weaken that part, almost delete another part." So the gate acts like a context-dependent filter.

The authors study five gate positions:

- G_1: after SDPA output,
- G_2: after value projection,
- G_3: after key projection,
- G_4: after query projection,

- G_5: after final dense output.

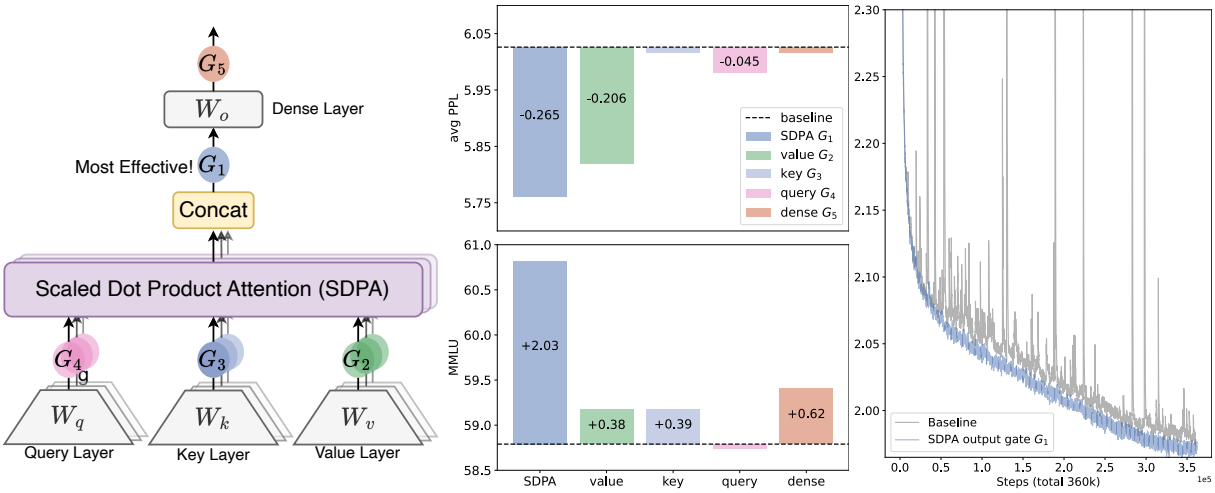

图 1: Original Figure 1 from the paper. Left: where the gate can be inserted. Middle: performance comparison for different gate positions on a 15B MoE model. Right: training-loss comparison on a 1.7B dense model.

How to read this figure

The middle panel is the key ablation result: not every gate position is equally good. The best place is **after SDPA**. The right panel shows another important message: the gated model trains more smoothly, with fewer sudden loss spikes.

3.3 Why gates at G_1 and G_2 are special

The authors argue that the value projection W_V and the output projection W_O together behave like two consecutive linear maps. Since the head dimension d_k is smaller than d_{model}, this path is effectively a **low-rank mapping**.

They write the attention output for token i and head k as

$$o_i^k = \left(\sum_{j=0}^{i} S_{ij}^k \cdot X_j W_V^k \right) W_O^k = \sum_{j=0}^{i} S_{ij}^k \cdot X_j (W_V^k W_O^k).$$

The important idea is not the notation, but the structure:

- first apply W_V,
- then later apply W_O,
- in between, if everything stays linear, expressiveness is limited.

So if we insert a non-linear operation between them, the mapping becomes more expressive. This motivates two useful choices:

$$o_i^k = \left(\sum_{j=0}^{i} S_{ij}^k \cdot \text{NonLinearity}(X_j W_V^k) \right) W_O^k,$$

$$o_i^k = \text{NonLinearity} \left(\sum_{j=0}^{i} S_{ij}^k \cdot X_j W_V^k \right) W_O^k.$$

These correspond roughly to:

- gating at G_2 (on values), and
- gating at G_1 (on SDPA output).

Grandma version of the non-linearity argument

Imagine two simple sieves placed one after another. If both are purely linear, the overall filtering power is limited. But if you insert a smart decision rule between them, the system can react differently to different inputs. That is what the gate adds.

4 Experimental Results

4.1 What models and data were used?

The paper studies both:

- **15B MoE models** with 2.54B activated parameters,
- **1.7B dense models**.

Training is done on subsets of up to **3.5 trillion tokens**. This is a very large-scale study, so the findings matter more than results from a tiny toy model.

Benchmarks include perplexity, MMLU, GSM8K, HumanEval, HellaSwag, C-Eval, and CMMLU.

4.2 Main ablation: where should the gate go?

The first major table compares many gate variants. The strongest pattern is:

- gating after **SDPA** is best overall,
- gating after **value projection** is also helpful,
- gating after query, key, or final output is much less helpful.

Most important empirical pattern

Useful gate positions are exactly the ones that insert non-linearity *between* W_V and W_O. That strongly supports the paper's explanation.

The paper also finds:

- **head-specific** is better than head-shared,
- **multiplicative** is better than additive,
- **sigmoid** is better than SiLU in the best configuration,
- even **headwise** gating gives strong gains with very few extra parameters.

That last point is practically important: the improvement is not just coming from adding many more parameters.

4.3 Dense models: does the trick still work?

Yes. On 1.7B dense models, SDPA gating still improves perplexity and benchmark scores. More interestingly, it also improves training stability.

The paper compares different settings:

- normal learning rate,
- larger learning rate,
- deeper model,
- larger batch size,
- longer training up to 3.5T tokens.

The baseline often becomes unstable when the learning rate is pushed high. The gated model remains trainable and can even benefit from the larger learning rate.

Why this matters in practice

If a model architecture can safely use larger learning rates or larger batches, training can become faster or scale better. So the gain is not only "slightly better benchmark scores"; it may also improve the engineering process of training big LLMs.

5 Why Does It Work? The Paper's Main Explanations

5.1 Reason 1: non-linearity

The authors test whether adding non-linearity alone helps. They try RMSNorm, SiLU, additive gating, and related variants.

The conclusion is:

Non-linearity helps consistently.

But non-linearity alone does not explain everything, because some non-linear variants are better than others. So the paper looks for a second ingredient.

5.2 Reason 2: sparsity

The gate values produced by sigmoid are often very small, especially for the best SDPA-output gate. This means many parts of the attention output are strongly suppressed.

What is sparsity here?

Sparsity means "many values are near zero." In this paper, the gate often gives small numbers, so large parts of the SDPA output are damped down. You can think of it as the model saying: "Most of this attention output is not useful for this query, so I will keep only a small relevant part."

The paper shows four key sparsity findings:

- the best gate has the **lowest average gate score**, so it is the sparsest,
- head-shared gating is less sparse and performs worse,
- value gating is less sparse and performs worse than SDPA-output gating,
- a modified non-sparse sigmoid also performs worse.

This leads to the paper's stronger claim:

The best gains come from input-dependent, head-specific sparse gating.

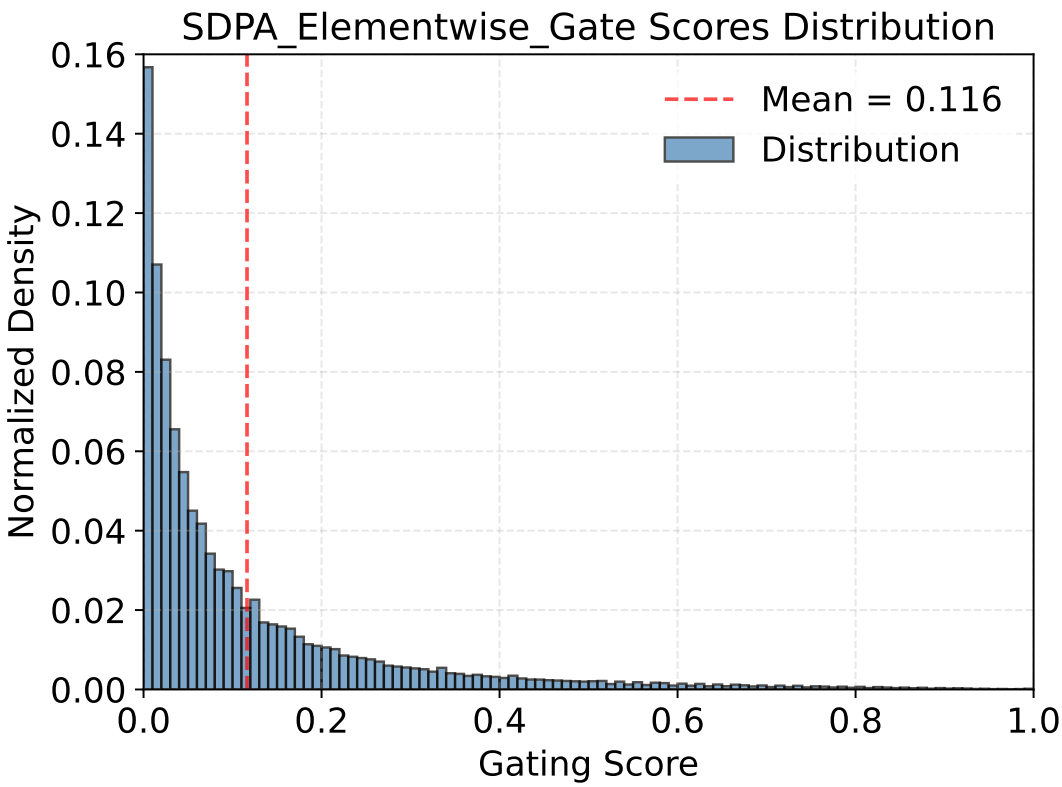

图 2: Original paper figure: distribution of SDPA elementwise gate values.

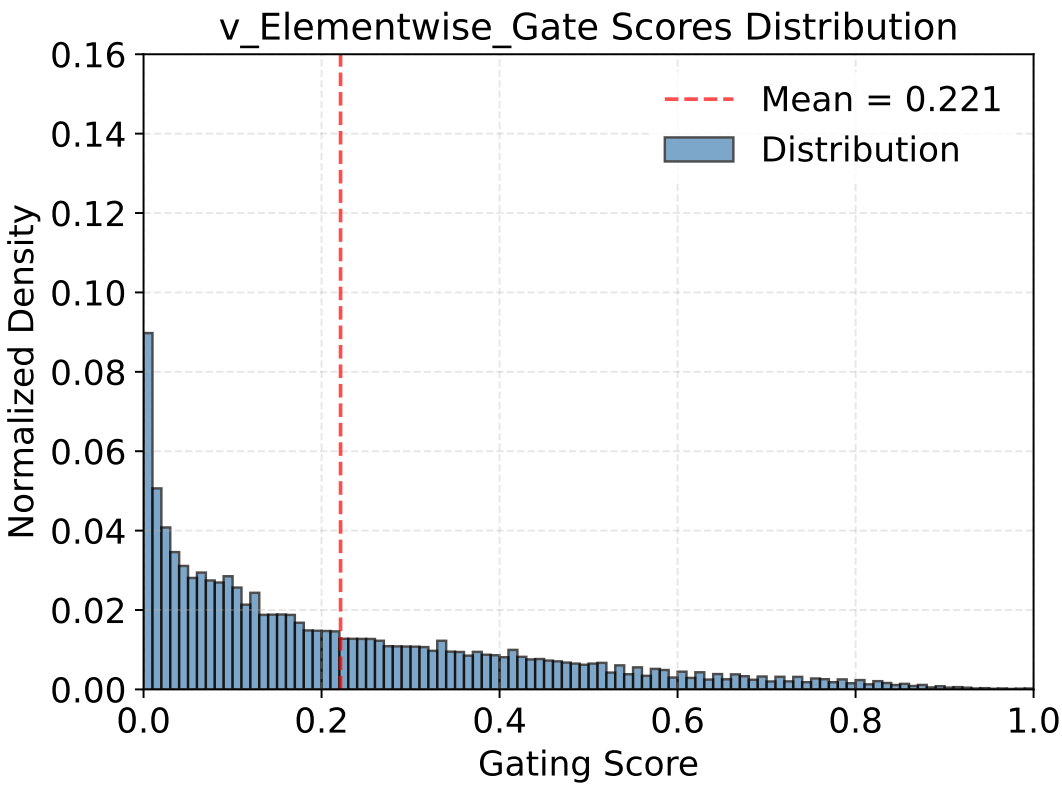

图 3: Original paper figure: distribution of value-gate scores.

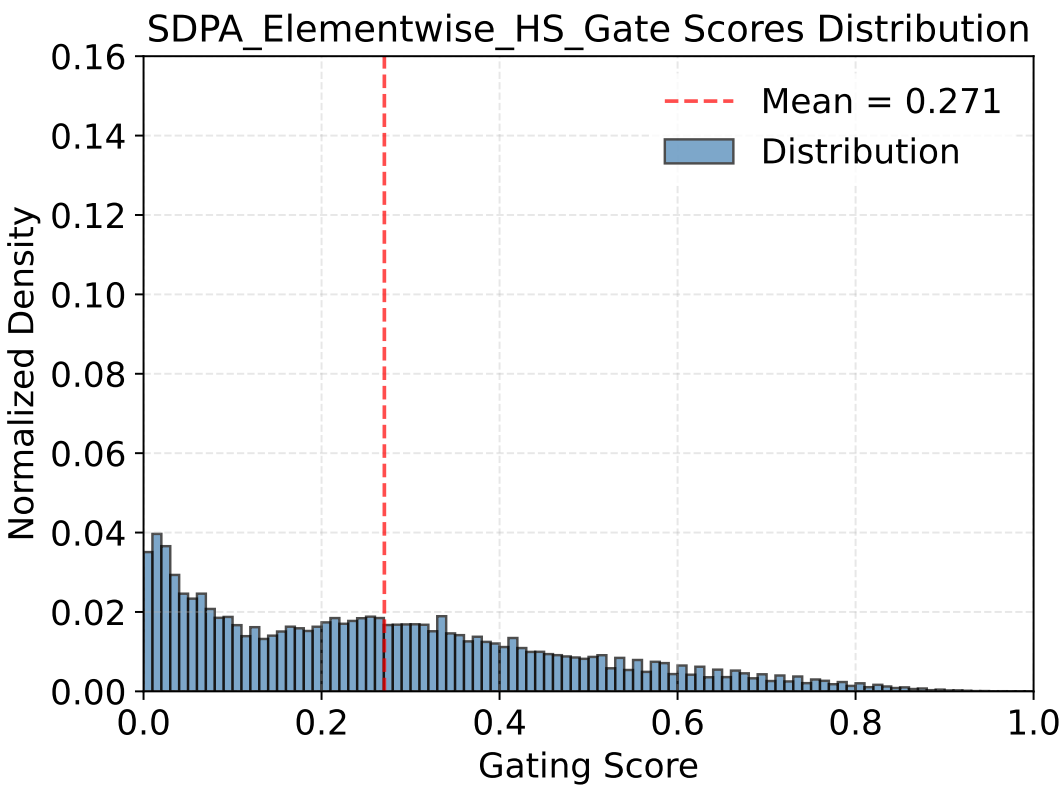

图 4: Original paper figure: distribution of SDPA gate values when heads share the gate.

What these distributions are telling us

The SDPA-output gate is concentrated much closer to zero than the others. In simple language: it is more willing to shut things off. The paper argues that this stronger, query-dependent filtering is one reason it works best.

5.3 Reason 3: less attention sink

One of the most interesting findings is that the gate almost removes the attention sink phenomenon.

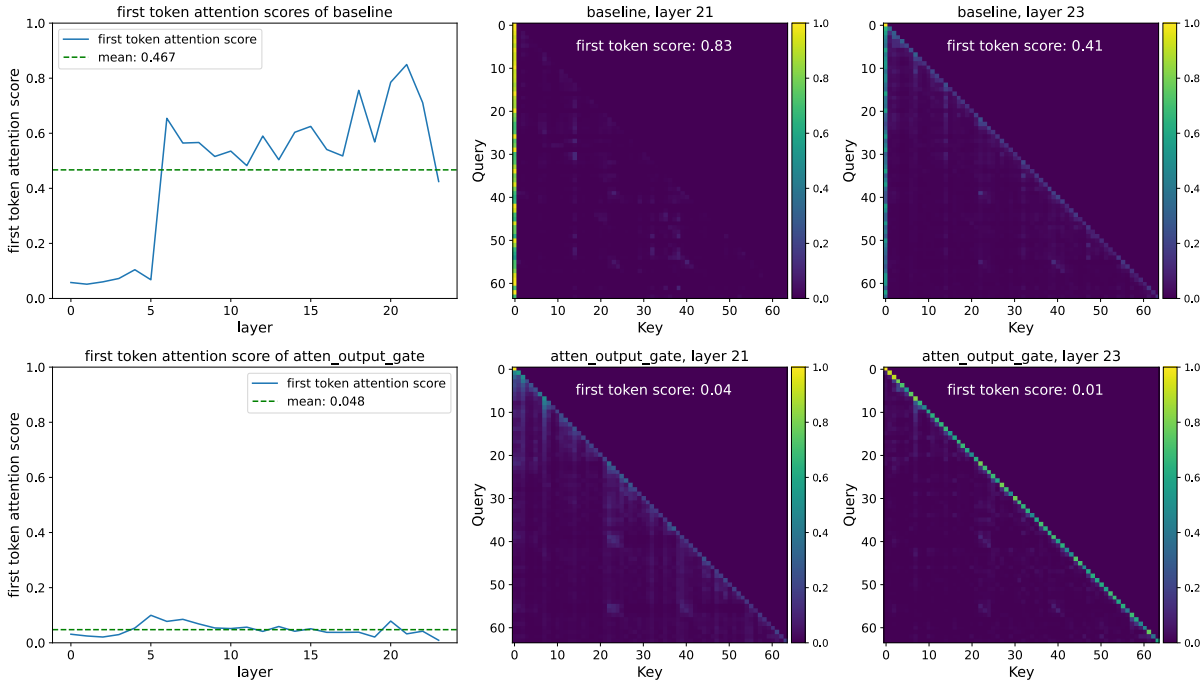

图 5: Original Figure 2 from the paper. Left: proportion of attention mass sent to the first token by layer. Right: average attention maps for a representative layer.

Reading the key number

In the baseline, the paper reports that on average 46.7% of attention scores go to the first token across layers. With the gate, that drops to 4.8%. That is a dramatic difference.

This matters because it suggests the gated model is not wasting as much attention on a token that acts like a default storage slot. Instead, attention becomes more content-sensitive.

The paper also studies **massive activations**, meaning unusually large hidden-state values. Gating reduces these too, especially in the best sparse configuration. The authors suggest this may explain better training stability, particularly under BF16 training, where very large activations can cause numerical problems.

Important subtlety

The paper also shows that reducing massive activations alone is *not* enough to eliminate attention sinks. For example, gating at the value layer can reduce activations but still leave some sink behavior. So the story is not just "smaller activations are good." The **query-dependent sparse filtering after SDPA** seems to be the crucial part.

6 Long-Context Results

The authors then ask: if attention sinks are reduced, does long-context behavior improve?

They extend models trained on 3.5T tokens to longer context lengths using RoPE-base modification and YaRN, then evaluate on RULER.

The main finding is:

- within trained context lengths, gating is slightly better,
- after extending to much longer contexts such as 64k and 128k, gating is **much better**.

Interpretation

The baseline may have partly learned to rely on the attention-sink pattern. When context extension changes the geometry of positions, that habit may become brittle. The gated model relies more on dynamic filtering and therefore adapts better.

This is not a formal proof, but it is a plausible and interesting hypothesis.

7 Extra Figures from the Appendix

The appendix contains several useful visual diagnostics. They help support the paper's mechanism story.

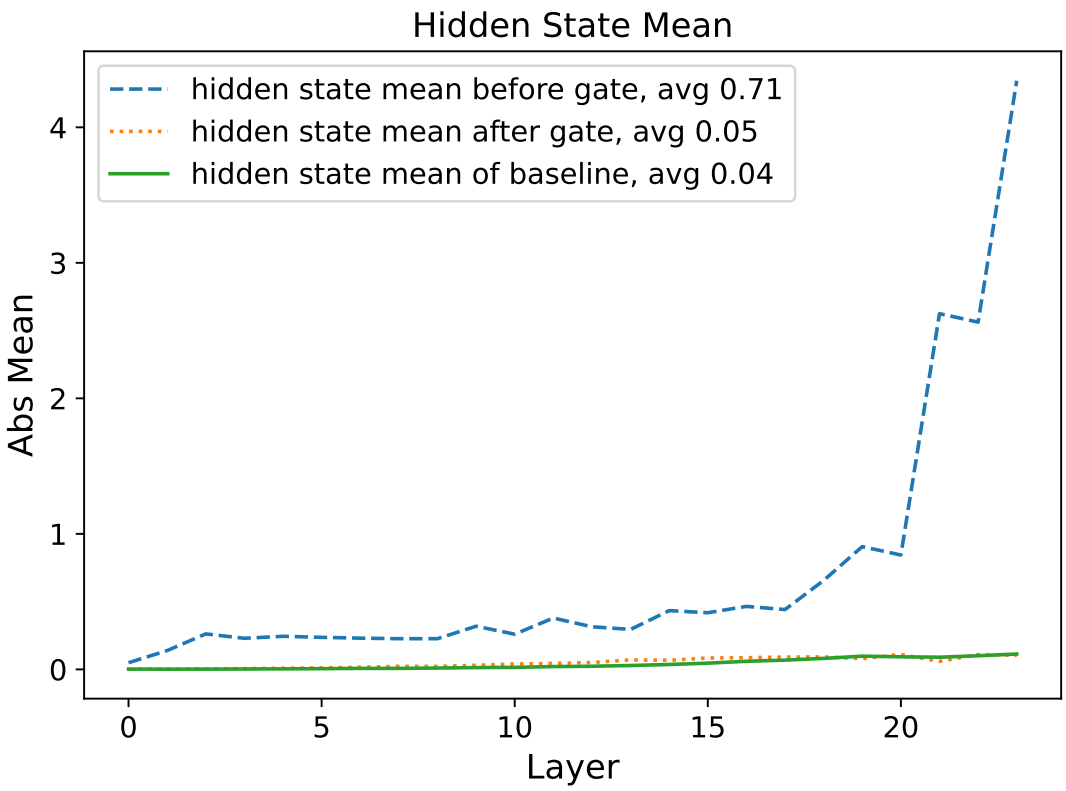

图 6: Original appendix figure: mean absolute values before and after gating.

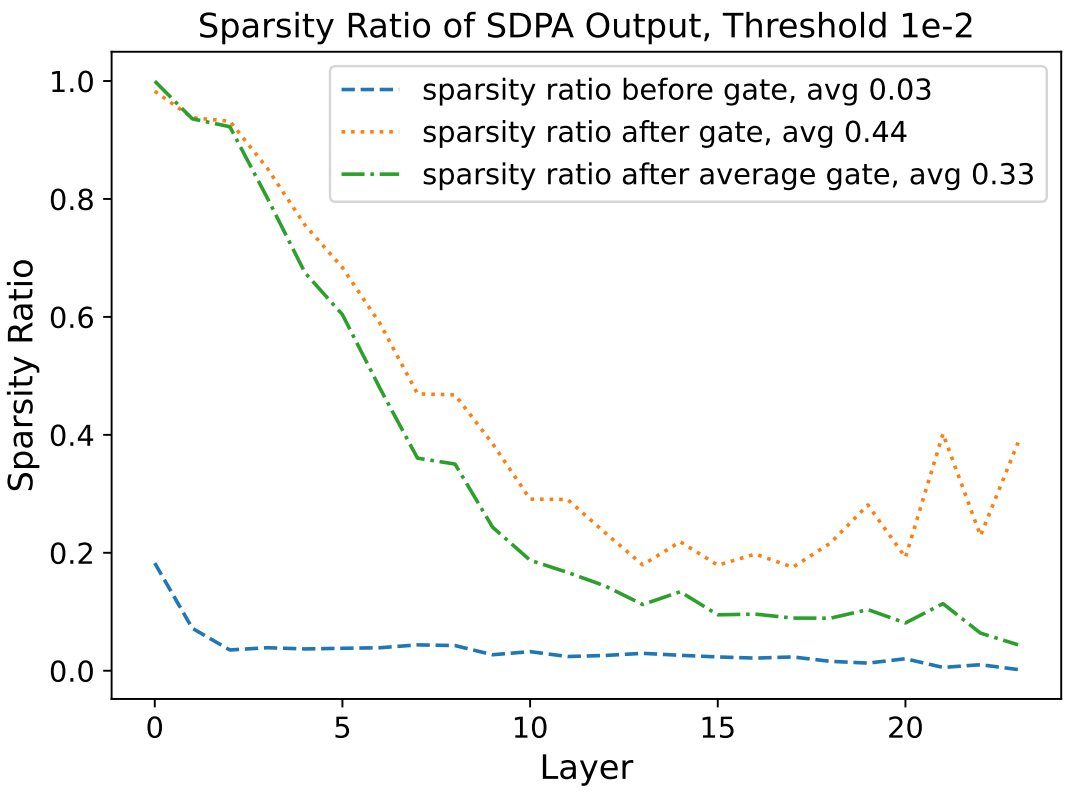

图 7: Original appendix figure: hidden-state sparsity under threshold 10^{-2}.

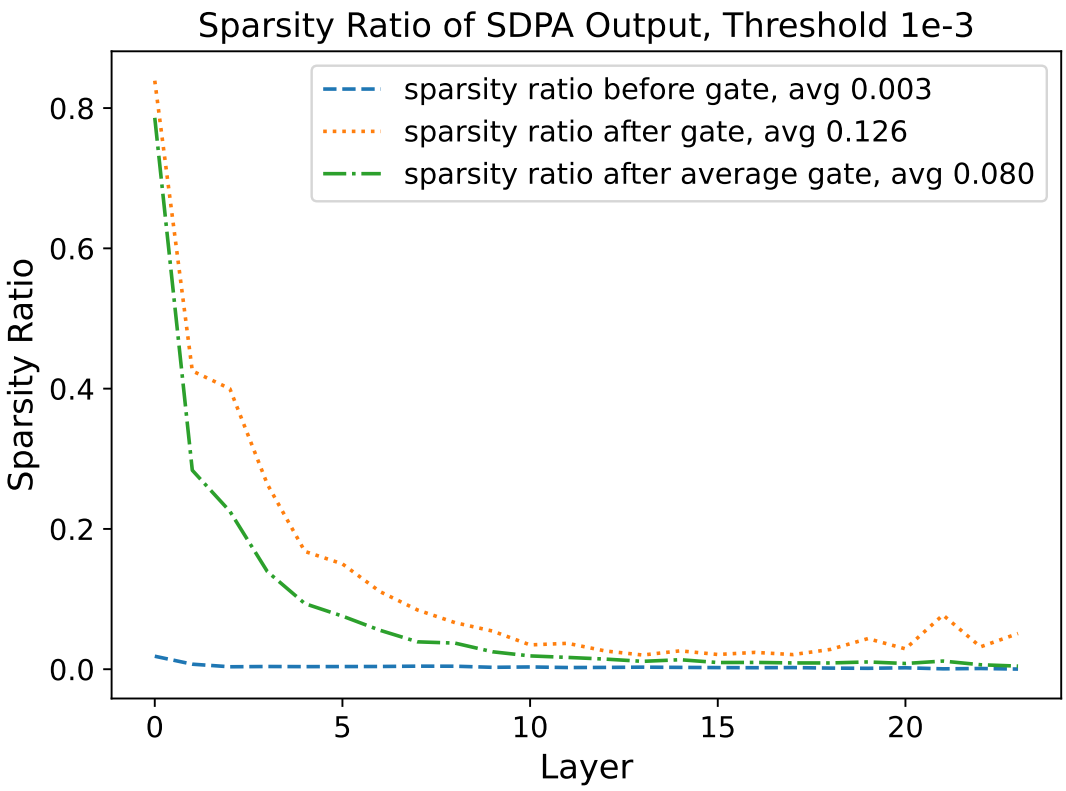

图 8: Original appendix figure: hidden-state sparsity under threshold 10^{-3}.

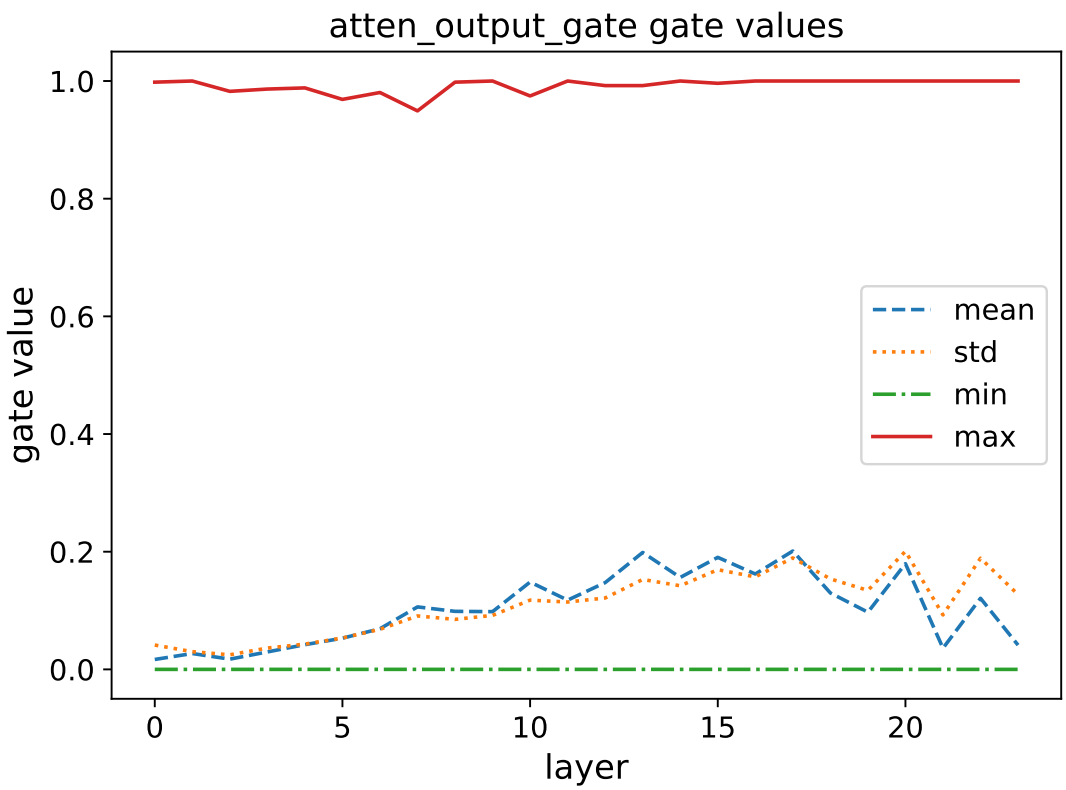

图 9: Original appendix figure: layerwise SDPA gate values.

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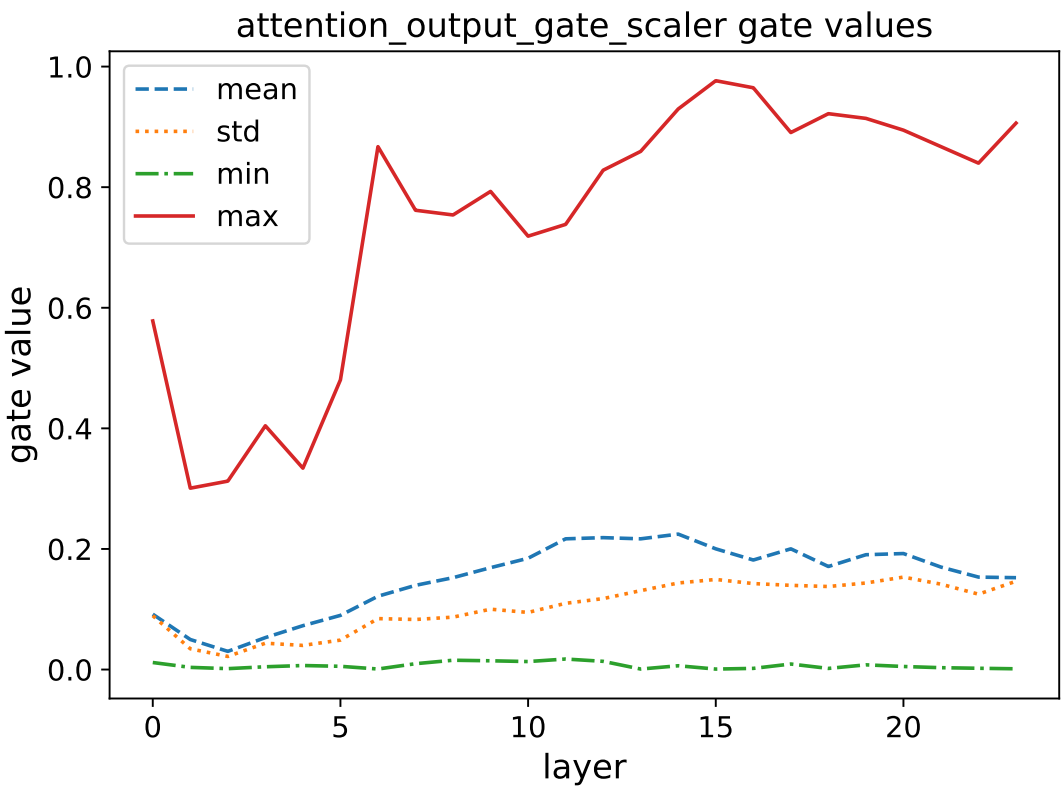

图 10: Original appendix figure: layerwise values for headwise SDPA gating.

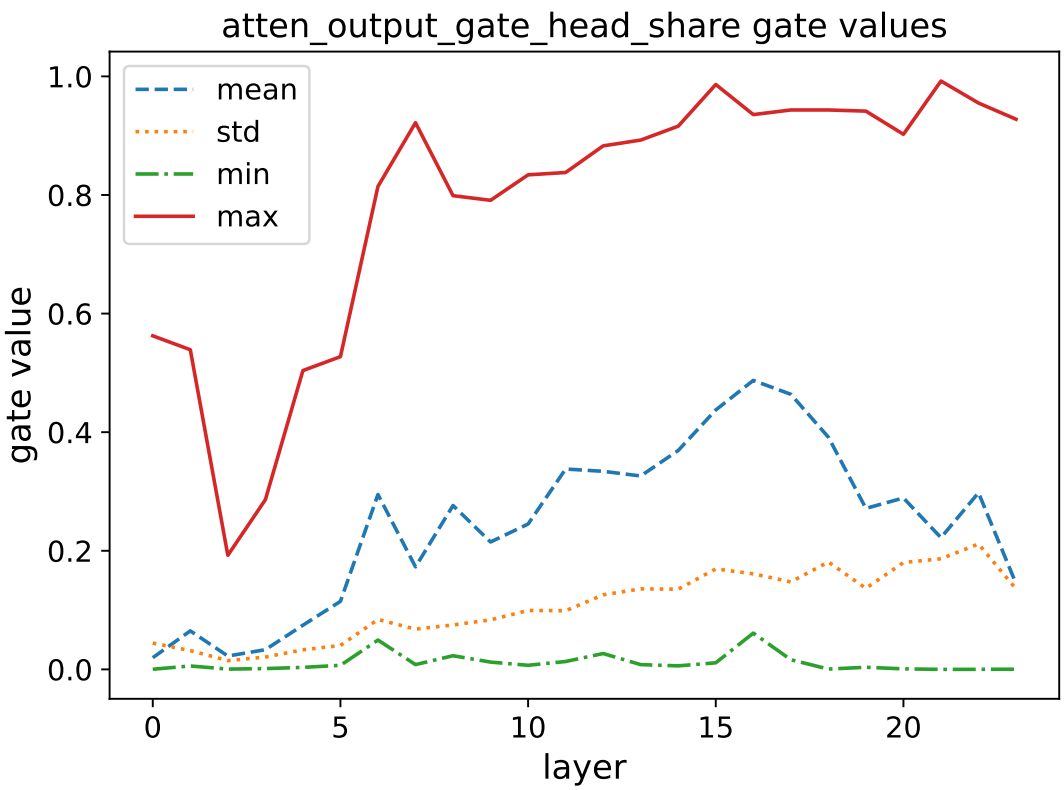

图 11: Original appendix figure: layerwise gate values with head-shared gating.

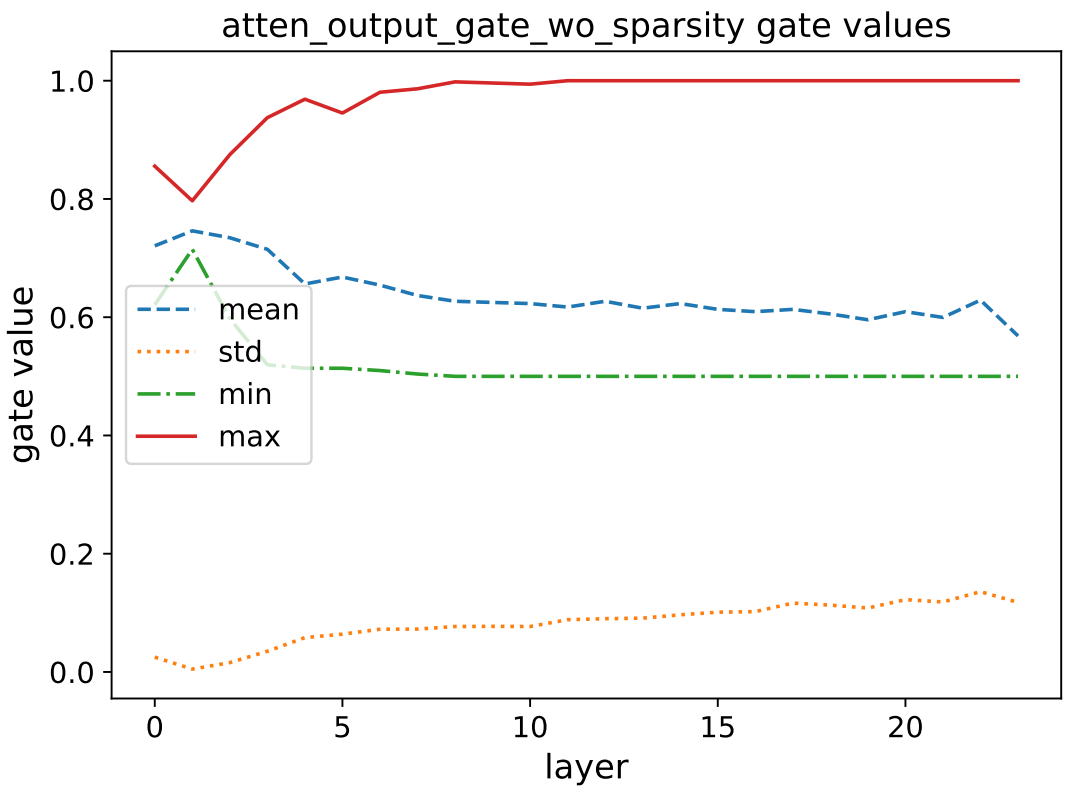

图 12: Original appendix figure: layerwise gate values when sparsity is weakened.

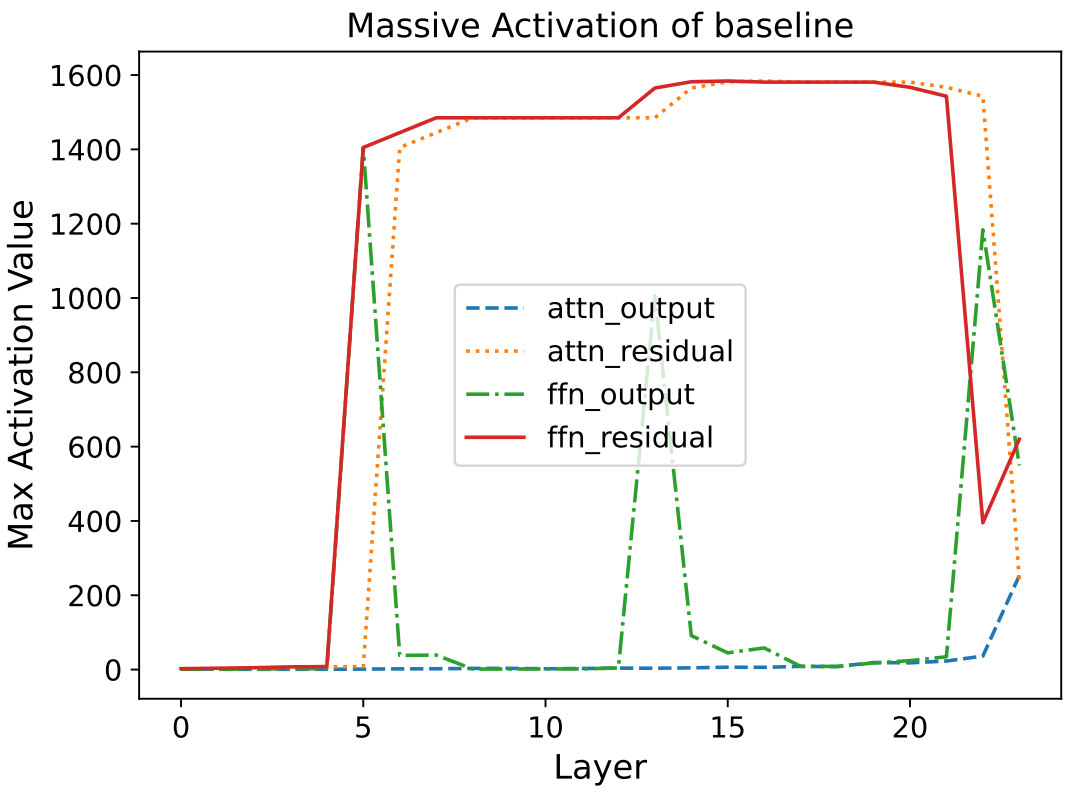

图 13: Original appendix figure: baseline massive activations across layers.

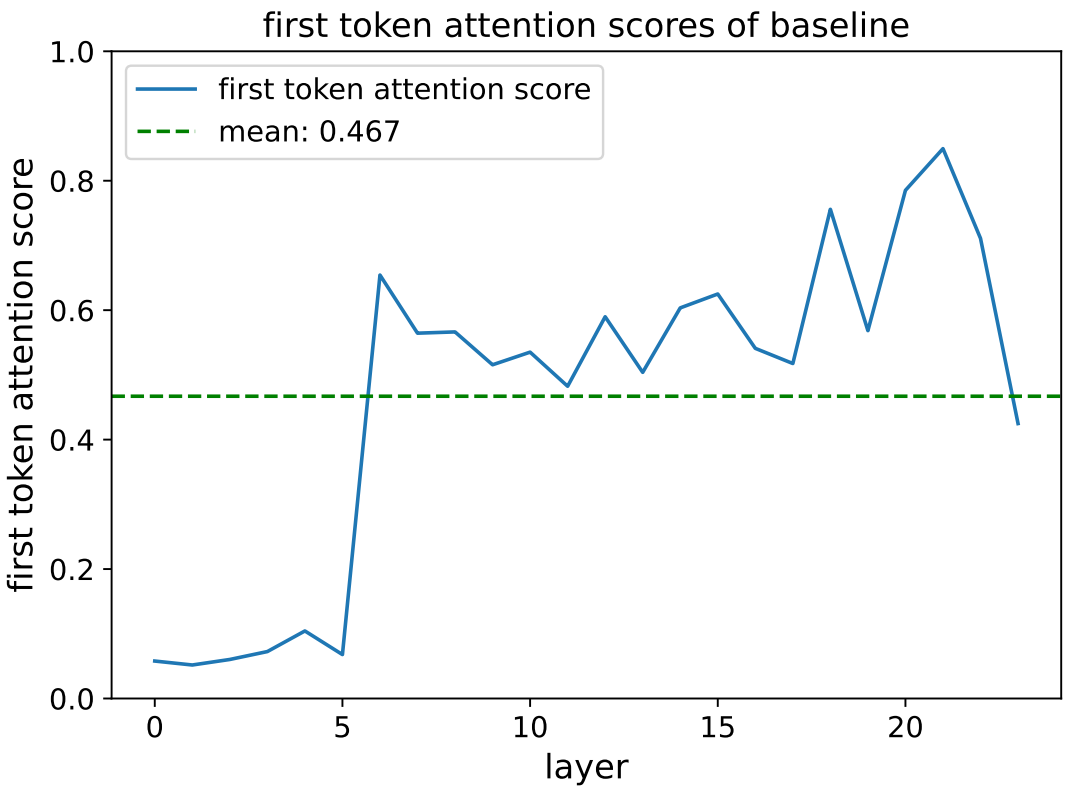

图 14: Original appendix figure: baseline attention-sink score on the first token across layers.

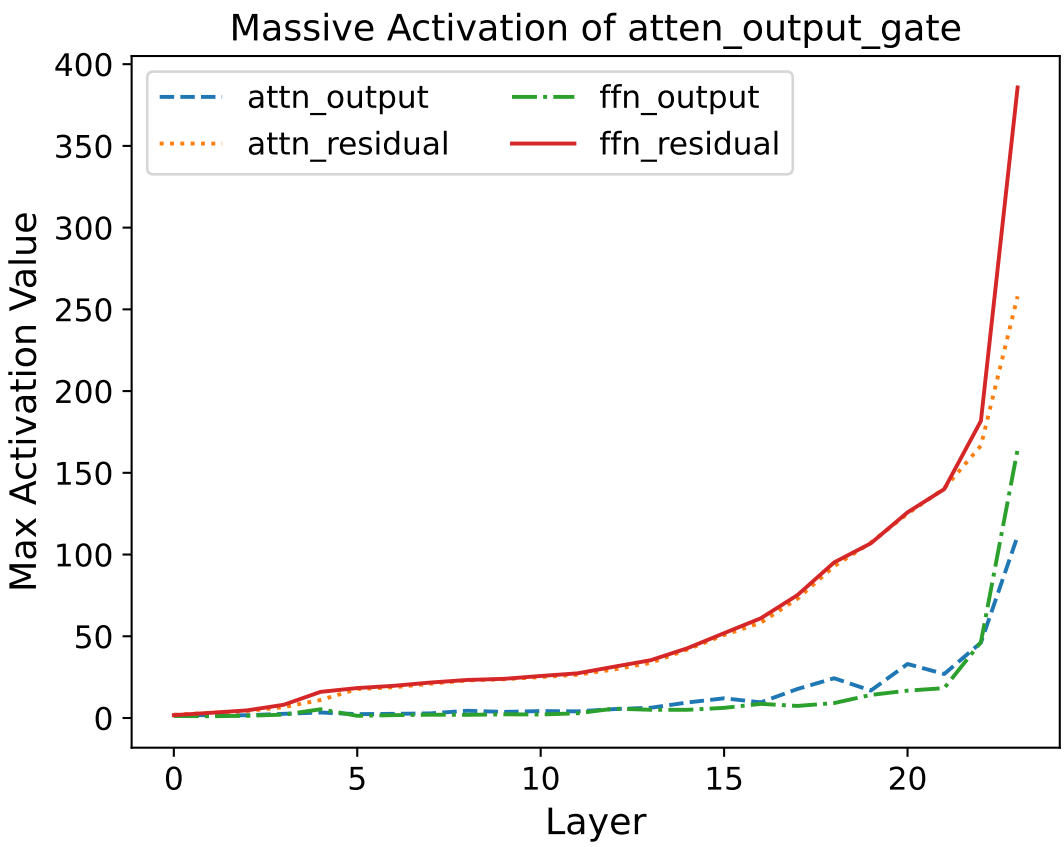

图 15: Original appendix figure: massive activations with SDPA gating.

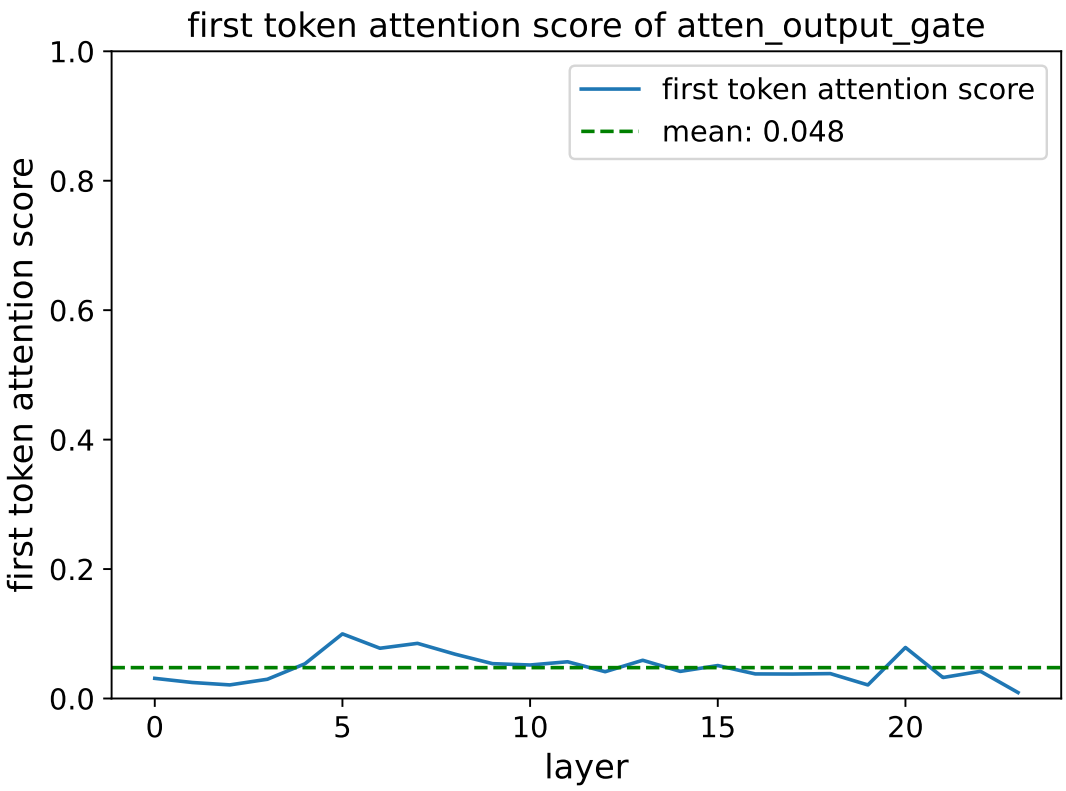

图 16: Original appendix figure: first-token attention with SDPA gating.

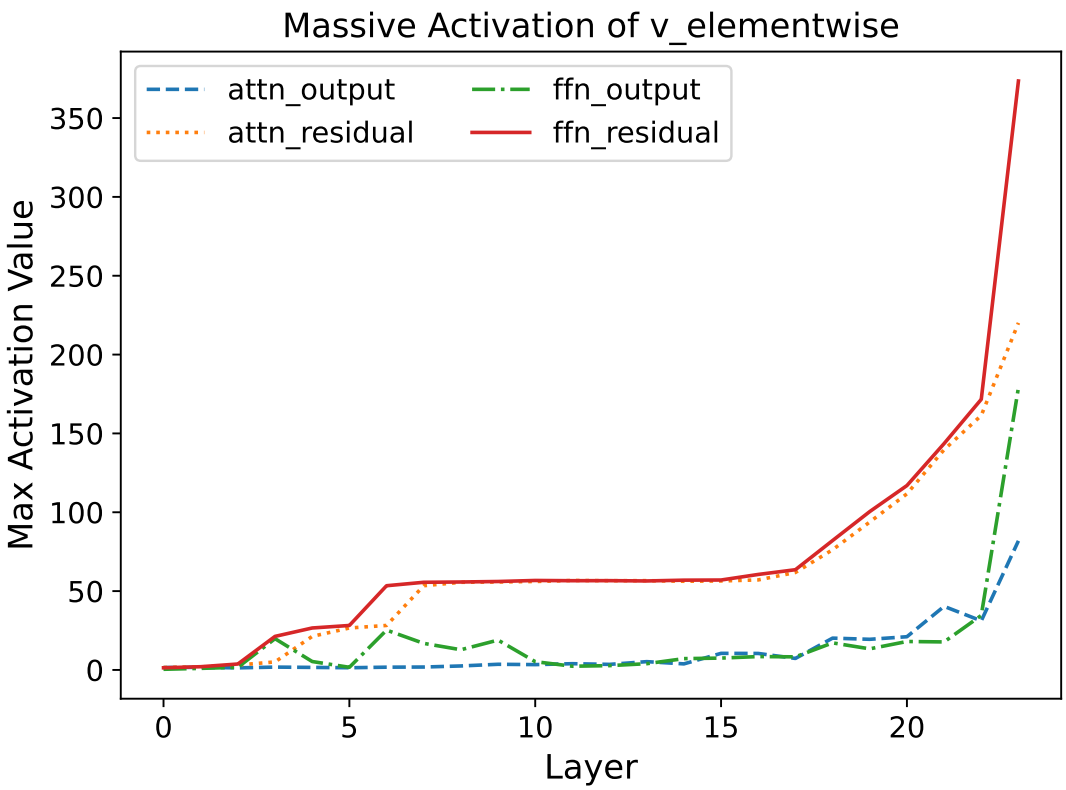

图 17: Original appendix figure: massive activations with value gating.

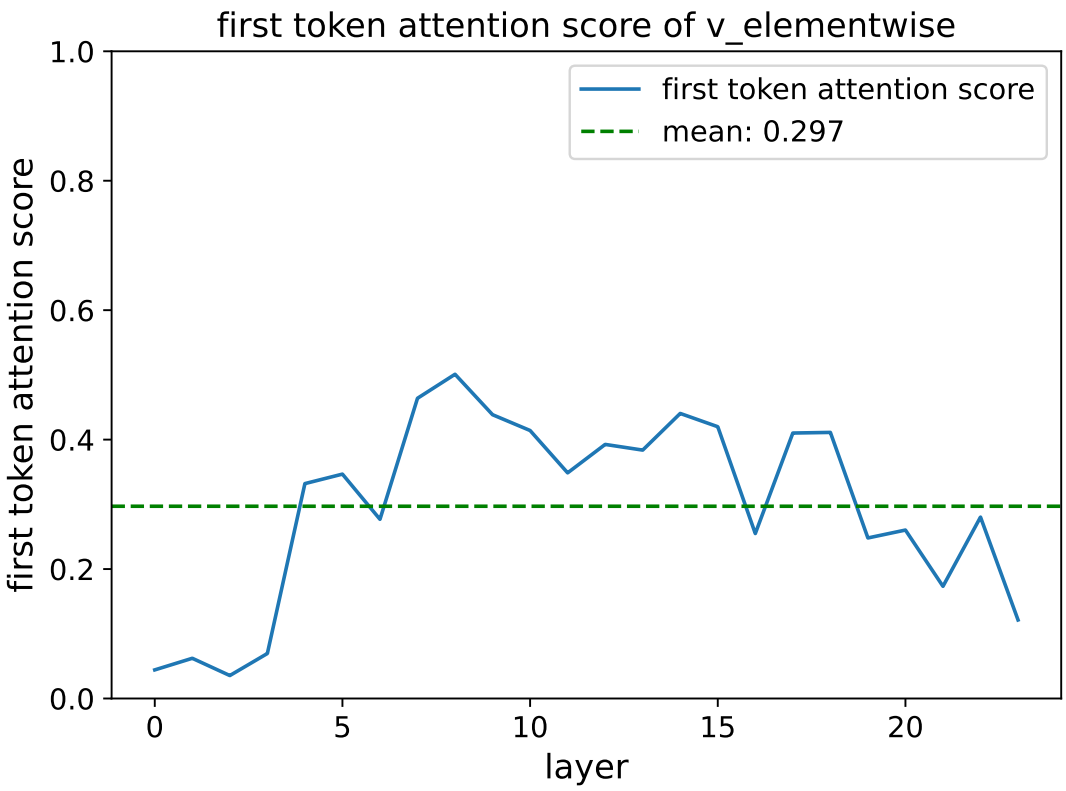

图 18: Original appendix figure: first-token attention with value gating.

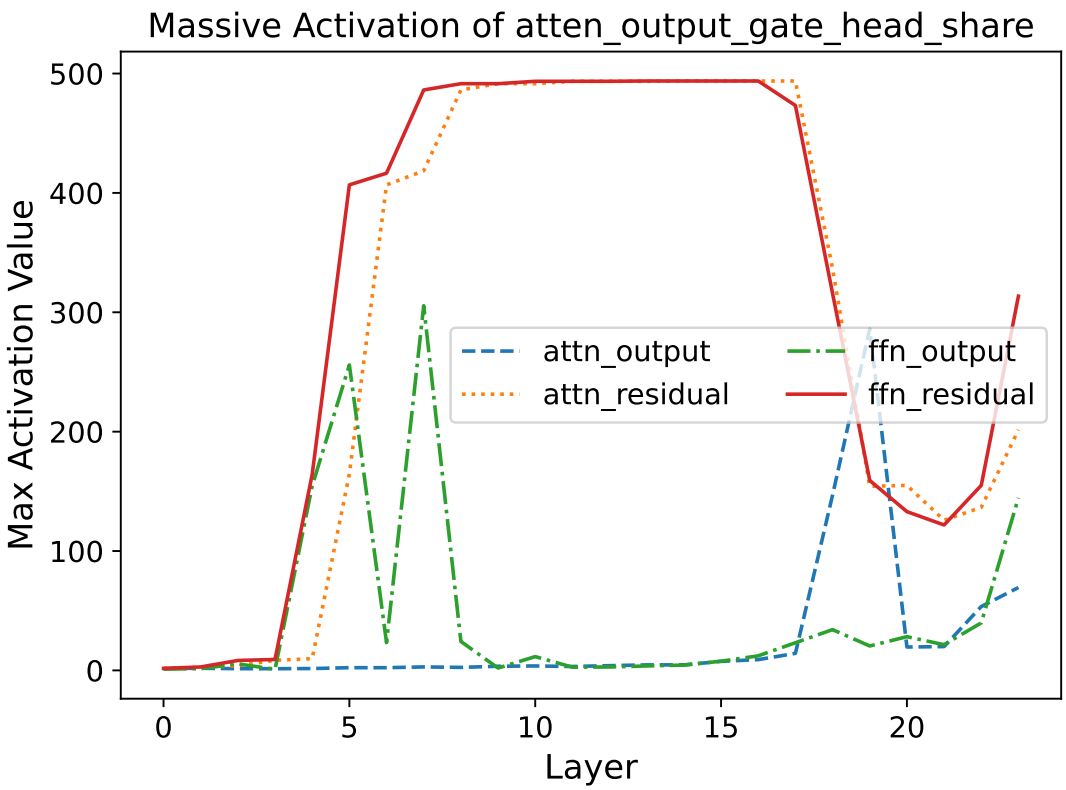

图 19: Original appendix figure: massive activations with head-shared SDPA gating.

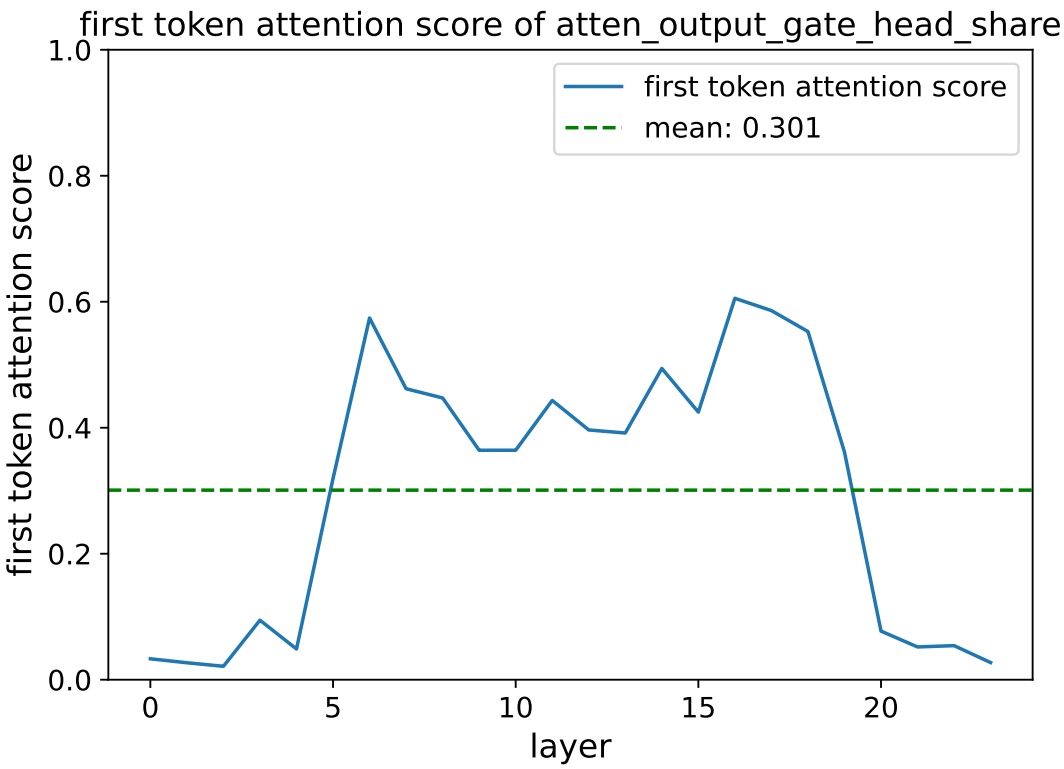

图 20: Original appendix figure: first-token attention with head-shared SDPA gating.

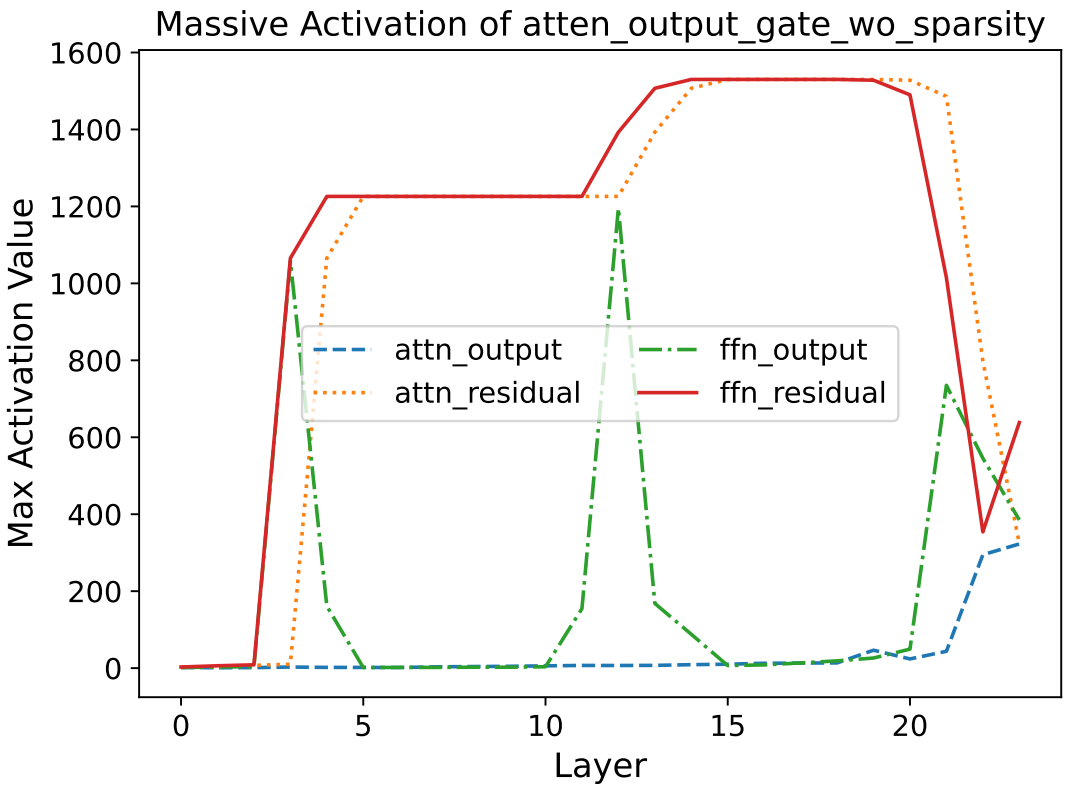

图 21: Original appendix figure: massive activations when gating sparsity is weakened.

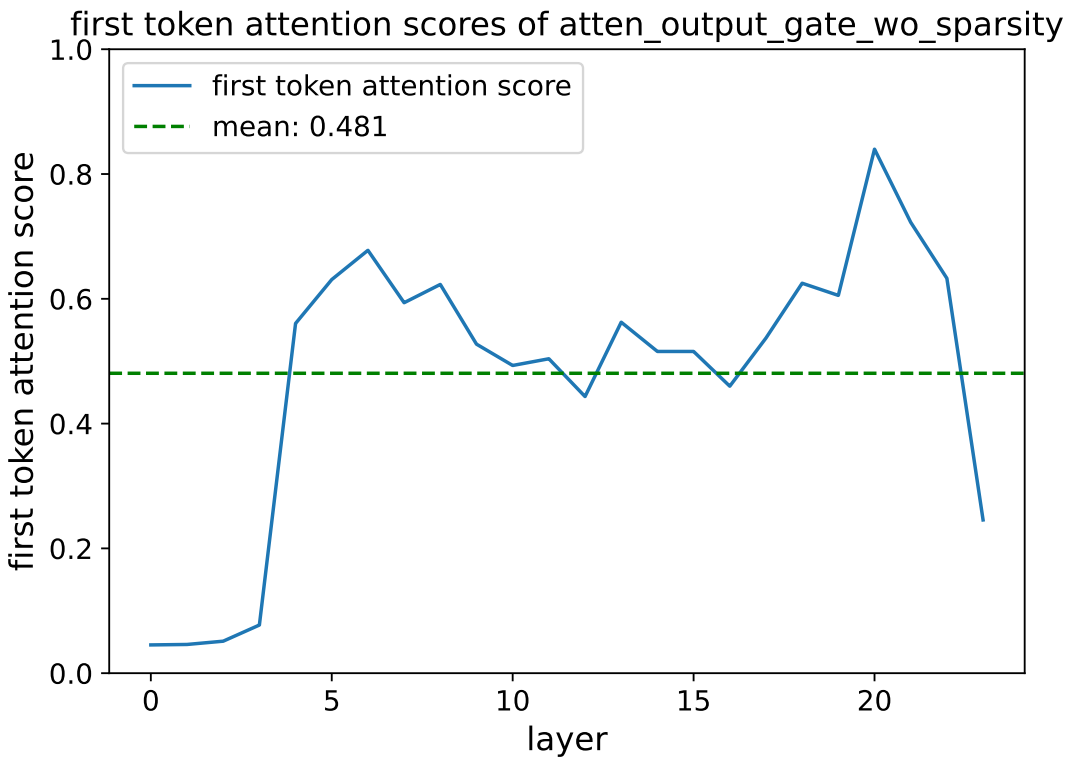

图 22: Original appendix figure: first-token attention when gating sparsity is weakened.

8 What Should a Beginner Remember?

Five takeaways from this paper

1. A very small architectural change inside attention can matter a lot.
2. The best place for the gate is **after SDPA output**, not everywhere.
3. The gain seems to come from **two ingredients together**: non-linearity and sparse filtering.
4. The best gate is **query-dependent and head-specific**, meaning different heads can filter information differently for different tokens.
5. This gate strongly reduces **attention sinks**, improves stability, and helps long-context extension.

9 Strengths, Limitations, and My Reading

9.1 Strengths

- **Very systematic ablation study.** The paper compares many nearby design choices.

- **Large-scale evidence.** Results are shown on both dense and MoE models, up to 3.5T tokens.
- **Mechanistic analysis.** The paper does not stop at "it works"; it tries to explain why.
- **Practical relevance.** Training stability and context extension are valuable in real LLM development.

9.2 Limitations

- The explanation is still mainly empirical, not a full theory.
- The long-context interpretation is plausible but not mathematically settled.
- The paper studies one family of gated softmax attention designs; there may be other equally strong designs not tested.

My overall judgment

This is a strong empirical paper. Its biggest contribution is not just a new trick, but a clearer picture of *what kind of gating helps attention and why*. Even if future architectures change, the lessons about non-linearity, sparsity, and attention sinks are likely to remain useful.

10 Reference

Citation

Zihan Qiu, Zekun Wang, Bo Zheng, Zeyu Huang, Kaiyue Wen, Songlin Yang, Rui Men, Le Yu, Fei Huang, Suozhi Huang, Dayiheng Liu, Jingren Zhou, and Junyang Lin. *Gated Attention for Large Language Models: Non-linearity, Sparsity, and Attention-Sink-Free*. arXiv:2505.06708.